Projective Systematic Authentication via Reed-Muller Codes

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Reference

[Liu and Mahdavifar(2024)] Hsuan-Po Liu and Hessam Mahdavifar. Projective systematic authentication via reed-muller codes, 2024. URL https://arxiv.org/abs/2404.09088.

System Structure of RM-A Authentication Scheme (1)

Entities:

- **Tx**:
  Given:
  - **Source**: $\mathbf{s} \in \mathbb{F}_2^M$
  - **Key**: $\mathbf{k} = (\mathbf{k}_1, \mathbf{k}_2)$, $\mathbf{k}_1 \in \mathcal{K}_1 \subset \mathbb{F}_2^n$, $\mathbf{k}_2 \in \mathcal{K}_2 = \mathbb{F}_2^l$

  Transmit signal:
  - **Encoder Scheme**: a generator matrix $\mathbf{G}$ of a RM(m, r), a $(n = 2^m, k = \sum_{i=0}^{r} \binom{m}{i})$ linear block code
  - **Encode**: $\mathbf{u_s} := [0, \ldots, 0, \mathbf{s}, 0] \in \mathbb{F}_2^k$, $\mathbf{c_s} = \mathbf{u_s}\mathbf{G}$
  - **Tag**: generated by $\mathbf{t} = \mathcal{E}_k(\mathbf{c_s})$
- **Receiver:** Holds the same shared key $\mathbf{k} = (\mathbf{k}_1, \mathbf{k}_2)$ and verifies authenticity.
- **Adversary:** Observes $(\mathbf{s}, \mathbf{t})$ and may attempt impersonation or substitution.

System Structure of RM-A Authentication Scheme (2)

Encoding Rule $\mathcal{E}_{\mathbf{k}} : \mathcal{C} \to \mathcal{T}$

1. $\mathbf{k}_1$ is designed to have weight $w_H(\mathbf{k}_1) = l$
2. According to $\mathbf{k}_1 \in \mathbb{F}_2^n$, selecting l entries from the codeword $\mathbf{c_s}$, $w_H(\mathbf{k}_1) = l$
3. $\mathbf{k}_2$ Masking: $\mathbf{t} = \mathbf{c}_{\mathbf{s},\mathbf{k}_1} + \mathbf{k}_2$.

System Structure of RM-A Authentication Scheme (3)

Transmission: Publicly send $(\mathbf{s}, \mathbf{t})$.
Verification:

1. Recompute $\mathbf{u} \rightarrow \mathbf{c_s} = \mathbf{uG}$ using the received $\mathbf{s}$.
2. Extract $\mathbf{c}_{\mathbf{s},\mathbf{k}_1}$ using $\mathbf{k}_1$.
3. Reconstruct tag: $\mathbf{t}' = \mathbf{c}_{\mathbf{s},\mathbf{k}_1} + \mathbf{k}_2$.
4. **Accept** if $\mathbf{t}' = \mathbf{t}$; otherwise, **reject**.

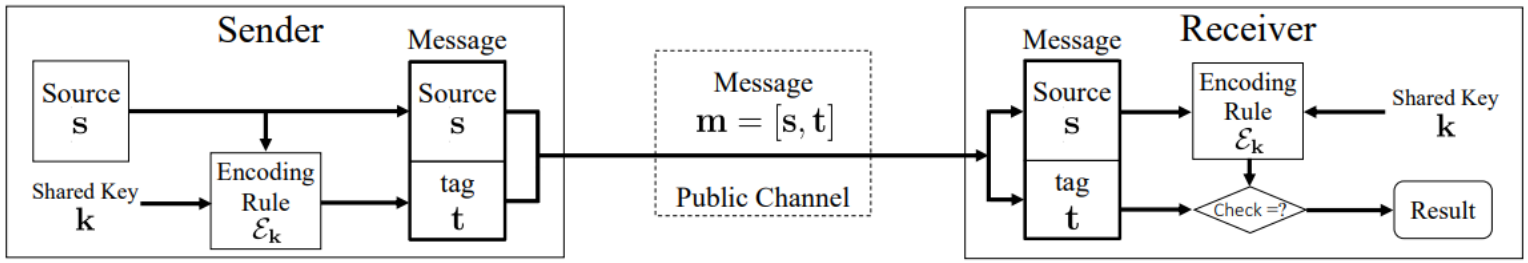
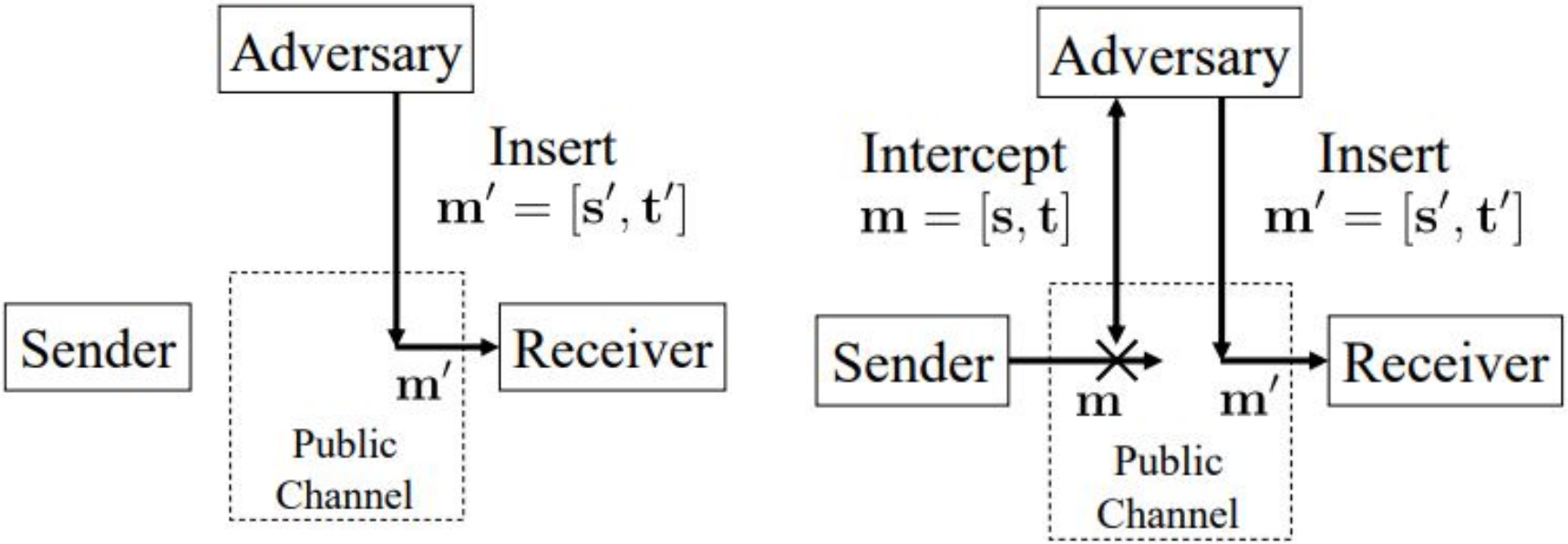


Figure: Algorithm2

Attack Models



(a) **Impersonation**: forge $[\mathbf{s}', \mathbf{t}']$ without observing communication

(b) **Substitution**: modify valid $[\mathbf{s}, \mathbf{t}]$ into forged $[\mathbf{s}', \mathbf{t}']$

Figure: Two types of Attack Models

Probability of Success Attacks

For the two models, the probability of success attacks are defined as

- **Impersonation**:

$$P_I := \max_{\mathbf{s}',\mathbf{t}'} \mathbb{P}\left([\mathbf{s}',\mathbf{t}'] \text{ valid}\right),$$

$$\mathbb{P}\left([\mathbf{s}',\mathbf{t}'] \text{ valid}\right) = \frac{|\{\mathbf{k} \in \mathcal{K} : \mathbf{t}' = \mathcal{E}_{\mathbf{k}}(\mathbf{s}')\}|}{|\{\mathbf{k} \in \mathcal{K}\}|}.$$

- **Substitution**:

$$P_S := \max_{\mathbf{s},\mathbf{t}} \max_{\mathbf{s}' \neq \mathbf{s},\mathbf{t}'} \mathbb{P}\left([\mathbf{s}',\mathbf{t}'] \text{ valid} \mid [\mathbf{s},\mathbf{t}] \text{ observed}\right),$$

$$\mathbb{P}\left([\mathbf{s}',\mathbf{t}'] \text{ valid} \mid [\mathbf{s},\mathbf{t}] \text{ observed}\right) = \frac{|\{\mathbf{k} \in \mathcal{K} : \mathbf{t} = \mathcal{E}_{\mathbf{k}}(\mathbf{s}),\ \mathbf{t}' = \mathcal{E}_{\mathbf{k}}(\mathbf{s}')\}|}{|\{\mathbf{k} \in \mathcal{K} : \mathbf{t} = \mathcal{E}_{\mathbf{k}}(\mathbf{s})\}|}.$$

P_I Analysis

Theorem 4.

$$P_1 = \max_{s',t'} \frac{|\{k \in \mathcal{K} : t' = \mathcal{E}_k(s') = c_{s',k_1+k_2}\}|}{|\{k \in \mathcal{K}\}|} = \frac{1}{2^l}.$$

Proof. Since $|\mathcal{K}_1| = \binom{n}{l}$ and $|\mathcal{K}_2| = 2^l$, we have

$$|\{k \in \mathcal{K}\}| = |\mathcal{K}| = |\mathcal{K}_1 \times \mathcal{K}_2| = \binom{n}{l} \cdot 2^l.$$

Since given $\mathbf{c_{s'}}, \mathbf{t'}$ and $\mathbf{k}_1$, there exist a unique $\mathbf{k}_2$, with $\mathcal{T} = \mathcal{K}_2$, we have

$$|\{\mathbf{k} \in \mathcal{K} : \mathbf{t'} = \mathcal{E}_{\mathbf{k}}(\mathbf{s'}) = \mathbf{c}_{\mathbf{s'},\mathbf{k}_1} + \mathbf{k}_2\}| = |\mathcal{K}_1| = \binom{n}{l}.$$

Thus $P_1 = \frac{1}{2^l}$. $\square$

P_S Expression (1)

Theorem

The quantity P_S, can be equivalently computed as

$$P_S = \max_{\tilde{\mathbf{s}} \neq 0_M} \max_{\tilde{\mathbf{t}}} \frac{\left|\left\{\mathbf{k}_1 \in \mathcal{K}_1 : \mathbf{c}_{\tilde{\mathbf{s}},\mathbf{k}_1} = \tilde{\mathbf{t}}\right\}\right|}{\binom{n}{\ell}},$$

where $\tilde{\mathbf{s}} \in \mathcal{S}$, excluding the all-zero source vector, and $\tilde{\mathbf{t}} \in \mathcal{T}$.

P_S Expression (2)

Proof. We have:

$$
\begin{aligned}
P_S &= \max_{\mathbf{s},\mathbf{t}} \max_{\mathbf{s}'\neq\mathbf{s},\mathbf{t}'} \mathbb{P}((\mathbf{s}',\mathbf{t}') \text{ valid} \mid \mathbf{s},\mathbf{t} \text{ observed}) \\
&= \max_{\mathbf{s},\mathbf{t}} \max_{\mathbf{s}'\neq\mathbf{s},\mathbf{t}'} \frac{\left|\left\{\mathbf{k}\in\mathcal{K} : \mathbf{t} = \mathbf{c}_{\mathbf{s},\mathbf{k}_1} + \mathbf{k}_2, \mathbf{t}' = \mathbf{c}_{\mathbf{s}',\mathbf{k}_1} + \mathbf{k}_2\right\}\right|}{\left|\left\{\mathbf{k}\in\mathcal{K} : \mathbf{t} = \mathbf{c}_{\mathbf{s},\mathbf{k}_1} + \mathbf{k}_2\right\}\right|} \\
&= \max_{\mathbf{s},\mathbf{t}} \max_{\mathbf{s}'\neq\mathbf{s},\mathbf{t}'} \frac{\left|\left\{\mathbf{k}\in\mathcal{K} : \mathbf{t} = \mathbf{c}_{\mathbf{s},\mathbf{k}_1} + \mathbf{k}_2,\ \mathbf{t}+\mathbf{t}' = \mathbf{c}_{\mathbf{s}+\mathbf{s}',\mathbf{k}_1}\right\}\right|}{\binom{n}{\ell}} \\
&= \max_{\mathbf{s},\mathbf{t}} \max_{\mathbf{s}'\neq\mathbf{s},\mathbf{t}'} \frac{\left|\left\{\mathbf{k}_1\in\mathcal{K}_1 : \mathbf{t}+\mathbf{t}' = \mathbf{c}_{\mathbf{s}+\mathbf{s}',\mathbf{k}_1}\right\}\right|}{\binom{n}{\ell}} \\
&= \max_{\mathbf{s},\mathbf{t}} \max_{\mathbf{s}'\neq\mathbf{s},\tilde{\mathbf{t}}} \frac{\left|\left\{\mathbf{k}_1\in\mathcal{K}_1 : \tilde{\mathbf{t}} = \mathbf{c}_{\mathbf{s},\mathbf{k}_1}\right\}\right|}{\binom{n}{\ell}} \\
&= \max_{\tilde{\mathbf{s}}\neq 0_M} \max_{\tilde{\mathbf{t}}} \frac{\left|\left\{\mathbf{k}_1\in\mathcal{K}_1 : \tilde{\mathbf{t}} = \mathbf{c}_{\tilde{\mathbf{s}},\mathbf{k}_1}\right\}\right|}{\binom{n}{\ell}}
\end{aligned}
$$

Lemma 9

The following inspect the equivalent problem of finding P_s by observing the upper bound of the set $\left\{\mathbf{k}_1 \in \mathcal{K}_1 : \tilde{\mathbf{t}} = \mathbf{c}_{\tilde{\mathbf{s}},\mathbf{k}_1}\right\}$.

Lemma (Upper Bound of $P_t(\mathbf{c}_{\tilde{\mathbf{s}}})$)

Consider the codeword $\mathbf{c}^{\star}_{\tilde{\mathbf{s}}} = [\mathbf{1}_w, \mathbf{0}_{n-w}]$.
For all codewords $\mathbf{c}'_{\tilde{\mathbf{s}}}$ *with the same weight* $w = w_H(\mathbf{c}'_s)$, *we have,*

$$P_t(\mathbf{c}^{\star}_{\tilde{\mathbf{s}}}) \geq P_t(\mathbf{c}'_{\tilde{\mathbf{s}}})$$

where $P_t(\mathbf{c}^{\star}_{\tilde{\mathbf{s}}})$ *is defined as*

$$P_t(\mathbf{c}_{\tilde{\mathbf{s}}}) = \max_{\tilde{\mathbf{t}}} \frac{\left|\left\{\mathbf{k}_1 \in \mathcal{K}_1 : \tilde{\mathbf{t}} = \mathbf{c}_{\tilde{\mathbf{s}},\mathbf{k}_1}\right\}\right|}{\binom{n}{\ell}}$$

Proof of Lemma 9 (1)

I don't like their proof so I used some different definitions. But the overall idea is the same.

Proof:

- **Case 1:** $\mathbf{c}_{\tilde{\mathbf{s}}} = \mathbf{c}^{\star}_{\tilde{\mathbf{s}}}$
  Given a valid w_t such that $w_t \leq w, w_t \leq l$, there exist exacly one valid tag $\tilde{\mathbf{t}}$ with $w_H(\tilde{\mathbf{t}}) = w_t$, which is,

$$\tilde{\mathbf{t}}^{\star} = [\mathbf{1}_{w_t}, \mathbf{0}_{l-w_t}]$$

  Given $\mathbf{c}^{\star}_{\tilde{\mathbf{s}}}, \tilde{\mathbf{t}}^{\star}$, the corresponding set of $\mathbf{k}_1$ has size

$$\left|\left\{\mathbf{k}_1 \in \mathcal{K}_1 : \tilde{\mathbf{t}}^{\star} = \mathbf{c}^{\star}_{\tilde{\mathbf{s}},\mathbf{k}_1}\right\}\right| = \binom{w}{w_t}\binom{n-w}{l-w_t}$$

  This is Lemma 7 of the paper.

Proof of Lemma 9 (2)

- **Case 2:** $\mathbf{c}_{\tilde{\mathbf{s}}} = \mathbf{c}'_{\tilde{\mathbf{s}}}$
  Let $\mathbf{c}'_{\tilde{\mathbf{s}}}$ be an arbitrary codeword $\in \mathcal{C}$ that has the same weight $\mathbf{w}_H(\mathbf{c}'_{\tilde{\mathbf{s}}}) = w$ Given a weight of the tag w_t, there may exists multiple $\tilde{\mathbf{t}}'$ that can be generated by the codeword. We define this set,

$$\mathcal{T}_{w_t} := \left\{\mathbf{t} \in \mathcal{T} : \mathbf{k}_1 \in \mathcal{K}, \mathbf{t} = \mathbf{c}'_{\tilde{\mathbf{s}},\mathbf{k}_1}, w_H(\mathbf{t}) = w_t\right\}$$

  the corresponding set of $\mathbf{k}_1$ that generates $\mathcal{T}_{w_t}$

$$\mathcal{K}_{1,w_t} := \left\{\mathbf{k}_1 \in \mathcal{K}_1 : \mathbf{t} \in \mathcal{T}_{w_t}, \mathbf{t} = \mathbf{c}'_{\tilde{\mathbf{s}},\mathbf{k}_1}\right\}$$

  We have exactly (lemma 6)

$$|\mathcal{K}_{1,w_t}| = \binom{w}{w_t}\binom{n-w}{l-w_t}$$

  Given $\mathbf{c}'_{\tilde{s}}$, The mapping $\mathcal{K}_{1,w_t} \to \mathcal{T}_{w_t}$ is surjective.

Proof of Lemma 9 (3)

For each $t \in \mathcal{T}_{w_t}$, $\mathcal{K}_{1,w_t}$ is partitioned into $|\mathcal{T}_{w_t}|$ subsets. Therefore for all $(\mathbf{c}'_{\tilde{s}}, \mathbf{t}' \in \mathcal{T}_{w_t})$ pairs we have

$$\left|\left\{\mathbf{k}_1 \in \mathcal{K}_1 : \tilde{\mathbf{t}}' = \mathbf{c}_{\tilde{\mathbf{s}}',\mathbf{k}_1}\right\}\right| \leq |\mathcal{K}_{1,w_t}| = \binom{w}{w_t}\binom{n-w}{l-w_t}$$

Thus we proved Lemma 9

$$P_t(\mathbf{c}^{\star}_{\tilde{\mathbf{s}}}) \geq P_t(\mathbf{c}'_{\tilde{\mathbf{s}}}) \quad \square$$

P_s Final Expression

Theorem (P_s Expression with only Weight)

$$P_s = \max_{w} \max_{w_t} \frac{\binom{w}{w_t}\binom{n-w}{l-w_t}}{\binom{n}{l}}$$

Conclusion

$$P_1 = \frac{1}{2^l}$$

$$P_s = \max_{w} \max_{w_t} \frac{\binom{w}{w_t}\binom{n-w}{l-w_t}}{\binom{n}{l}}$$

Numerical Results

m	4	5	6	7	8
Size	$(16, 5)$	$(32, 6)$	$(64, 7)$	$(128, 8)$	$(256, 9)$
P_{I}	0.125	0.125	0.125	0.125	0.125
P_{S}	0.4	0.3817	0.3810	0.3780	0.3765

Figure: $M = 4, l = 3, r = 1$

Some Comments

- The paper assume attacks are random. We might be able to reverse back to obtain $\mathbf{k}$.
- Extending to other linear code
  - We still have the same upper bounds for P_s,
  - But tracking the weight distribution is harder. (But does this even matter?)
- When will $P_s = \max_{\mathbf{c}_{\tilde{s}}} P_t(\mathbf{c}_{\tilde{s}}) = P_t(\mathbf{c}_{\tilde{s}})$

Possible Extension

- SOTA of authentication system designs
- Lower bound of the Probability of success attacks, or other common performance metrics
- Justification of adopting this system?